EEG Causal Graphs for Three-Class Classification

Preprocessing, directed connectivity estimation, and baseline classification for AD, CN, and FTD

Michele Arrigoni

September 2026

Abstract

This project builds a graph-based representation of resting-state EEG recordings for three-class classification among Alzheimer's disease (AD), healthy controls (CN), and frontotemporal dementia (FTD). Each subject is represented by a graph whose nodes are the 19 EEG channels, whose node attributes describe local spectral and statistical activity, and whose edges encode directed connectivity estimated from time-series methods.

The pipeline combines filtering over a 0.5–40 Hz passband, resampling to 100 Hz, epoch quality control, and stationarity testing. Spectral features are computed before differencing, while connectivity is estimated using Granger causality, VARLiNGAM, and PCMCI. Each method produces separate directed and symmetric graph representations. The current exports contain 87 OpenNeuro and 85 BrainLat subjects. Connectivity is summarized separately for each dataset and clinical class. A compact MLP is trained and evaluated independently within each dataset. On OpenNeuro, VARLiNGAM achieves the highest observed test balanced accuracy (0.605). On BrainLat, Granger and VARLiNGAM tie for the highest observed test balanced accuracy (0.821). These single-split results do not measure cross-dataset generalization.

Contents

1 Objective and Data Representation **3**

1.1 Project objective . . . 3

1.2 Processed datasets . . . 3

2 Preprocessing Pipeline **4**

2.1 Filtering, resampling, and epochs . . . 4

2.2 Quality control . . . 4

2.3 Stationarity handling . . . 4

3 Node Features **5**

4 Directed Connectivity Estimation **5**

4.1 Common representation and temporal aggregation . . . 5

4.2 Granger causality . . . 6

4.3 VARLiNGAM . . . 6

4.4 PCMCI . . . 7

5 Connectivity Results **7**

5.1 OpenNeuro 7

5.2 BrainLat 10

6 Three-Class MLP Classification **14**

6.1 Experimental setup 14

6.2 OpenNeuro: independent within-dataset evaluation 14

6.3 BrainLat: independent within-dataset evaluation 15

6.4 Interpretation and scope 16

7 Output Artifacts **17**

8 Limitations and Future Work **17**

9 Conclusion **18**

1 Objective and Data Representation

1.1 Project objective

The objective is to transform multichannel EEG recordings into homogeneous graph samples suitable for graph neural models and for simpler subject-level classification baselines. For each subject s, the recording is represented as a graph:

$$G_s = (V_s, X_s, A_s, y_s),$$

where V_s is the fixed set of EEG electrodes, X_s contains node features, A_s is a weighted adjacency matrix, and $y_s \in \{0, 1, 2\}$ is the clinical label. Label 0 denotes CN, label 1 denotes AD, and label 2 denotes FTD.

The term causal is used in the statistical sense of the adopted estimators: Granger, VARLiNGAM, and PCMCI provide evidence that the past or structural state of one channel helps explain another channel under the model assumptions. In this report, an edge therefore means a model-based directional dependence in the observed EEG channels. It does not prove a direct physiological pathway, because scalp EEG is affected by volume conduction, reference choices, latent neural sources, and estimator assumptions.

1.2 Processed datasets

The current processed exports contain 87 OpenNeuro subjects and 85 BrainLat subjects. OpenNeuro subject `sub-086` and BrainLat subject `sub-078` are excluded from the analyses because their preprocessing outputs are not compatible with the shared-duration exports used for the report. All connectivity summaries and MLP experiments in this report are computed separately by dataset. The 172 analyzed subjects are not pooled for classification because OpenNeuro and BrainLat differ in cohort composition, acquisition conditions, retained duration, and electrode geometry; pooling them would make dataset identity a potential confounder.

Dataset	CN	AD	FTD	Total	Epochs/subject
OpenNeuro	29	36	22	87	61
BrainLat	32	35	18	85	28

Table 1: Current processed exports, analyzed separately by dataset.

BrainLat electrodes are mapped to the 19 canonical channels by selecting available electrodes nearest to normalized standard-1020 montage coordinates, without reusing an electrode. This harmonizes channel names and order but does not establish identical acquisition geometry across datasets. The coordinates are used only to choose the closest available BrainLat sensor for each target 10–20 label; they are not used as spatial features in the classifier.

All graphs share the same 19-channel order:

Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2,
F7, F8, T3, T4, T5, T6, Fz, Cz, Pz.

This order is enforced before feature extraction and adjacency construction, so the same row and column always refer to the same electrode. Equivalently, every feature vector and adjacency matrix can be read as if it used the same fixed electrode layout.

2 Preprocessing Pipeline

2.1 Filtering, resampling, and epochs

Recordings are loaded from EEGLAB files with MNE, restricted to the canonical 19 channels, band-pass filtered between 0.5 and 40 Hz, notch-filtered at 50 Hz, and resampled to 100 Hz. The Nyquist frequency is the highest frequency that can be represented after sampling without aliasing, and is one half of the sampling rate:

$$f_N = \frac{100}{2} = 50 \text{ Hz},$$

which preserves the delta, theta, alpha, beta, and low-gamma ranges used by the node features.

The continuous signal is split into non-overlapping 5-second epochs:

$$L = f_s T = 100 \cdot 5 = 500$$

samples per epoch before stationarity differencing. OpenNeuro outputs are cropped to 61 valid epochs per subject in the current processed dataset, while BrainLat outputs are cropped to 28 valid epochs per subject. Within each dataset the pipeline takes the first shared-minimum number of accepted epochs, corresponding to 305 and 140 seconds of selected data respectively before differencing; these windows need not be contiguous in the original recording because rejected epochs are skipped. This choice gives all subjects within a dataset the same number of epochs, preventing recording duration from becoming an unintended classifier signal.

2.2 Quality control

Epochs are discarded when they contain non-finite values, extreme amplitudes, or nearly flat channels. The amplitude threshold is:

$$150 \times 10^{-6} \text{ V},$$

which corresponds to 150 microvolts in the unit convention normally returned by MNE. Amplitude screening is performed before channel standardization.

Accepted epochs are standardized channel by channel:

$$z_{i,t} = \frac{x_{i,t} - \mu_i}{\sigma_i + \varepsilon}.$$

Here $x_{i,t}$ is the sample of channel i at time t, μ_i and σ_i are the mean and standard deviation of that channel within the epoch, and ε is a small constant that prevents division by zero. This normalization reduces scale differences across channels and subjects before causal estimation.

2.3 Stationarity handling

Granger-style autoregressive inference assumes approximately stationary time series. Each channel is tested using ADF and KPSS:

$$p_{\text{ADF}} \leq 0.05, \qquad p_{\text{KPSS}} > 0.05.$$

When one or more channels fail the criterion, the epoch is differenced along time:

$$\Delta X_t = X_t - X_{t-1},$$

up to two times. Differenced epochs are used for causal estimation. At the limit the epoch is retained even if it still fails the tests, to avoid discarding too much data after quality control. The stationarity checks are therefore a preprocessing heuristic rather than a guarantee that every final series is perfectly stationary.

Spectral node features are computed from standardized epochs before differencing. This preserves low-frequency content, particularly in the delta band, which can be distorted by the high-pass effect of differencing. The pipeline retains separate pre-differencing and stationarity-processed representations for feature extraction and connectivity estimation, respectively.

3 Node Features

For each channel, the feature vector has eight components:

$$x_i = [P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma, \mu, \sigma, \sigma^2].$$

The five spectral features are band powers of the Z-scored signals estimated with Welch's method over:

$$\delta : 0.5\text{–}4, \quad \theta : 4\text{–}8, \quad \alpha : 8\text{–}13, \quad \beta : 13\text{–}24, \quad \gamma : 24\text{–}40 \text{ Hz}.$$

Mean, standard deviation, and variance are computed over the accepted pre-differencing samples of the channel.

Stacking the eight features for all 19 channels yields the subject-level node-feature matrix:

$$X_s \in \mathbb{R}^{19\times 8}.$$

4 Directed Connectivity Estimation

4.1 Common representation and temporal aggregation

Connectivity is estimated independently for each subject and epoch from the stationarity-processed signals. With $N = 19$ channels, each method produces an $N \times N$ directed matrix with $N(N-1) = 342$ possible off-diagonal connections. Throughout this section, $W_{ij}^{(e,m)}$ denotes the raw epoch-level weight from source j to target i for method m. Self-connections are excluded from the resulting connectivity matrices. Autoregressive self-history can nevertheless be used internally by the estimators when predicting a channel: a target channel may use its own past as a baseline predictor, but the saved graph does not store self-edges on the diagonal.

The subject-level matrix is formed as

$$W_s^{(m)} = \frac{1}{|\mathcal{E}_s^{(m)}|} \sum_{e\in\mathcal{E}_s^{(m)}} W_s^{(e,m)},$$

where $\mathcal{E}_s^{(m)}$ is the set of epochs included in the aggregation. Granger and VARLiNGAM average the supplied epochs; PCMCI excludes epochs whose estimation fails and returns a zero matrix if none succeeds. Consequently, a zero PCMCI output must be interpreted together with processing diagnostics. Averaging yields a static summary and removes information about when individual connections occur within the recording.

4.2 Granger causality

Pairwise Granger causality measures whether past values of a source channel improve prediction of a target channel beyond the target's own past:

$$Y_t = \sum_{r=1}^{p} a_r Y_{t-r} + \sum_{r=1}^{p} b_r X_{t-r} + \eta_t.$$

The null hypothesis is:

$$H_0 : b_1 = b_2 = \cdots = b_p = 0.$$

For each ordered pair of channels and each epoch, autoregressive orders $p \in \{1, 2, 3, 4\}$ are tested using the residual-sum-of-squares F test implemented by `statsmodels`. At 100 Hz, these lags correspond to 10–40 ms of past activity. This range is important because Granger causality can only detect predictive effects that occur within the tested delay window: short windows emphasize fast EEG interactions, while longer windows would increase model complexity and computational cost. Each order tests the joint contribution of all source lags from 1 to p, rather than testing only the individual lag p.

For every source–target pair, the implementation keeps the smallest finite p-value across the tested orders:

$$\pi_{ij}^{(e)} = \min_{p \in \{1,2,3,4\}} \pi_{ij,p}^{(e)}.$$

The candidate p-values are then corrected across all ordered pairs in the epoch using Benjamini–Hochberg false-discovery-rate control. The raw epoch-level weight is:

$$W_{ij}^{(e,G)} = \begin{cases} -\log_{10}\left(\max(\pi_{ij}^{(e)}, 10^{-12})\right), & \text{if the FDR-corrected test is significant,} \\ 0, & \text{otherwise.} \end{cases}$$

Subject-level Granger adjacency is the average across accepted epochs.

A smaller p-value gives a larger positive score after the $-\log_{10}(\cdot)$ transformation. This score measures statistical evidence against the no-predictive-effect null hypothesis; it is neither a probability of causation nor the magnitude of a regression coefficient. The analysis is pairwise, so it does not condition on the other 17 channels. Indirect paths and common drivers can therefore contribute to the detected relationships. The configured significance level is $\alpha = 0.01$.

4.3 VARLiNGAM

VARLiNGAM is a vector autoregressive extension of LiNGAM. It models multivariate time series and estimates instantaneous and lagged structural effects:

$$x(t) = B_0 x(t) + B_1 x(t-1) + \cdots + B_p x(t-p) + e(t).$$

The implementation configures a maximum order of four samples, uses BIC for order selection, and enables coefficient pruning. BIC penalizes overly complex autoregressive models, while pruning suppresses coefficients that the fitted VARLiNGAM model regards as weak. Let p_e denote the number of lag matrices returned for epoch e. The instantaneous matrix B_0 participates in the model but is excluded from the stored connectivity aggregation because lagged effects are easier to compare with Granger and PCMCI, which are also stored as positive-lag directed graphs. The lagged matrices are averaged in absolute value:

$$W_{ij}^{(e,L)} = \begin{cases} \dfrac{1}{p_e} \displaystyle\sum_{\tau=1}^{p_e} \left| B_{\tau,ij}^{(e)} \right|, & p_e > 0, \\ 0, & p_e = 0. \end{cases}$$

This produces a directed non-negative adjacency matrix in the same `target, source` orientation used by the other methods.

Unlike the Granger score, this weight summarizes fitted structural coefficient magnitudes in a multivariate model. Taking absolute values is appropriate here because the downstream graph representation uses non-negative edge strengths; it prevents opposite signs across lags from cancelling. The cost is that the saved graph no longer distinguishes excitatory-like and inhibitory-like signed effects. Averaging also removes the individual delay associated with each coefficient. Interpretation depends on the linear structural model and its assumptions, including independent non-Gaussian innovations and an acyclic instantaneous structure; the aggregated lagged graph is not required to be acyclic.

4.4 PCMCI

PCMCI identifies lagged dependencies in multivariate time series through a parent-selection stage followed by conditional-independence testing. The acronym refers to the combination of PC-style parent selection and momentary conditional independence testing. This project uses the ParCorr test. For a candidate source j and target i, PCMCI tests whether:

$$X_{t-\tau}^{(j)} \not\perp\!\!\!\perp X_t^{(i)} \mid \mathbf{Z},$$

where $\mathbf{Z}$ contains selected lagged parents. The notation reads as: "the source channel j at lag τ is not conditionally independent of the current target channel i, given the selected conditioning set $\mathbf{Z}$." The parent-selection stage reduces the conditioning sets, and the momentary conditional independence stage evaluates candidate links given the selected variables. With ParCorr, dependence is measured through the correlation of residuals after linear conditioning. The implementation uses analytic significance tests and sets both the parent-selection threshold and the final edge threshold to 0.01 through the shared project configuration.

For a source–target pair, define the significant positive-lag set as

$$\mathcal{T}_{ij}^{(e)} = \{\tau \in \{1, 2, 3, 4\} : p_{j,i,\tau}^{(e)} < 0.01\}.$$

The raw epoch-level weight is

$$W_{ij}^{(e,P)} = \begin{cases} \dfrac{1}{|\mathcal{T}_{ij}^{(e)}|} \displaystyle\sum_{\tau \in \mathcal{T}_{ij}^{(e)}} |\rho_{j,i,\tau}^{(e)}|, & |\mathcal{T}_{ij}^{(e)}| > 0, \\ 0, & \text{otherwise}, \end{cases}$$

where ρ is the partial-correlation value returned by PCMCI. Only significant lags enter this mean: a link with one strong significant lag can therefore have a larger weight than a link with several weaker ones. Lag-zero links and self-connections are excluded from the stored graph.

Multivariate conditioning addresses dependencies explained by the selected observed variables, but does not establish immunity to unobserved confounding. ParCorr targets linear conditional relationships, and taking absolute values discards their signs. The final filtering uses the returned p-values directly, without false-discovery-rate adjustment.

5 Connectivity Results

5.1 OpenNeuro

This dataset contains 87 subjects (29 CN, 36 AD and 22 FTD), with 61 retained epochs per subject.

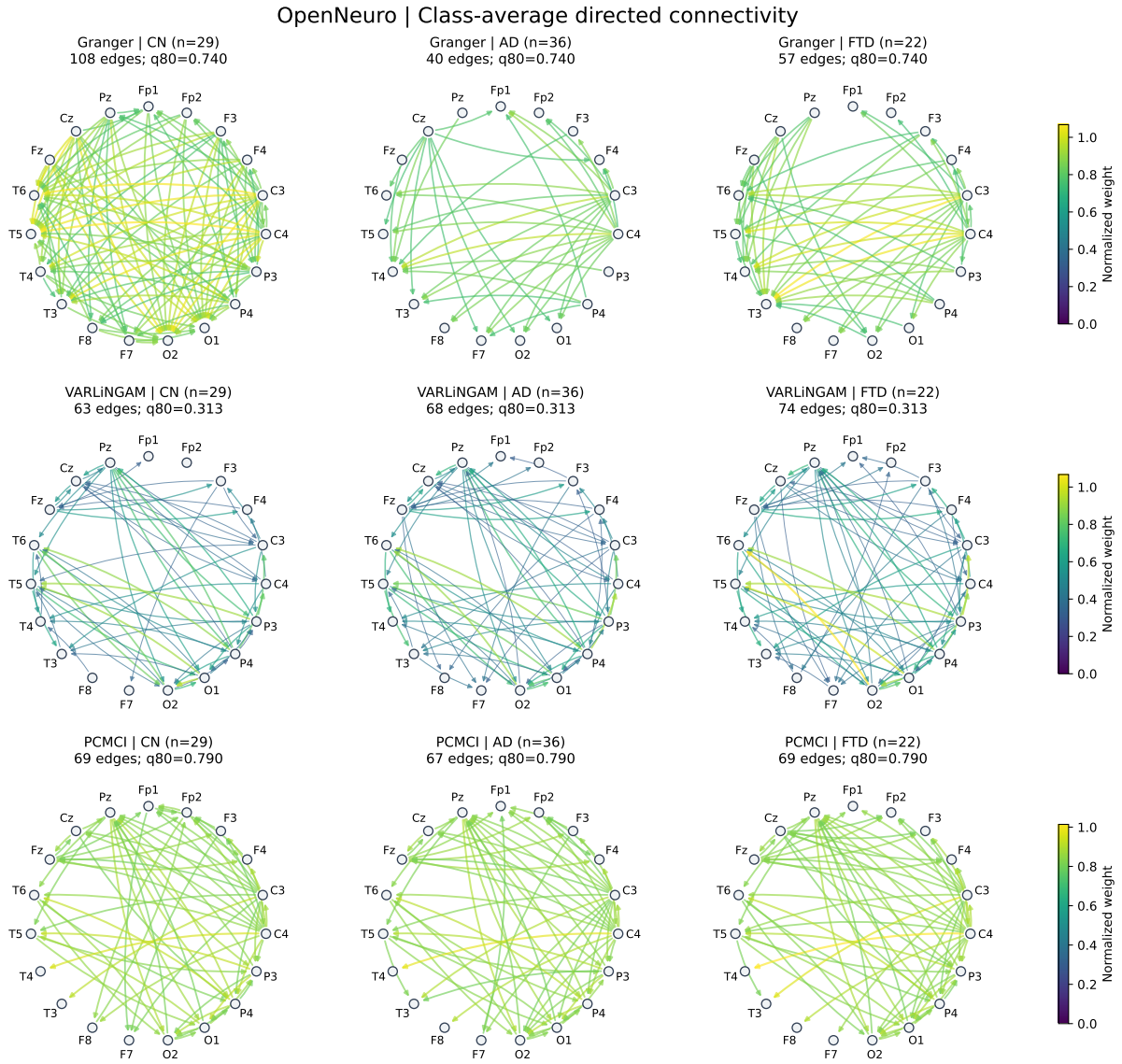


Figure 1: OpenNeuro: class-average directed networks. Arrows run from source to target; color and width encode normalized weight. Circular positions indicate channel order, not anatomy. Scales and thresholds are shared across classes within each method.

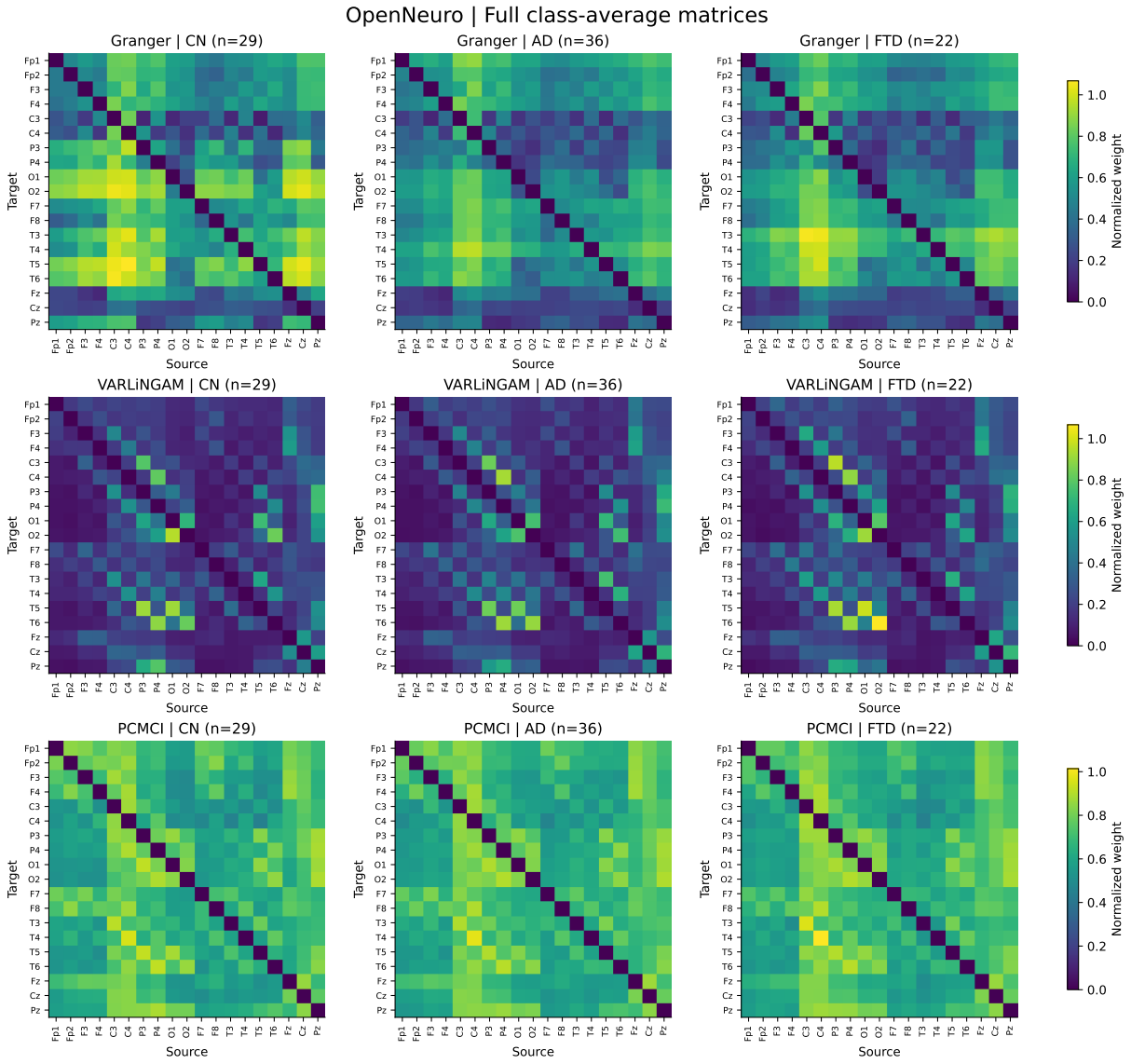


Figure 2: OpenNeuro: unthresholded class-average matrices. Rows are targets, columns are sources; each method shares its color scale across classes.

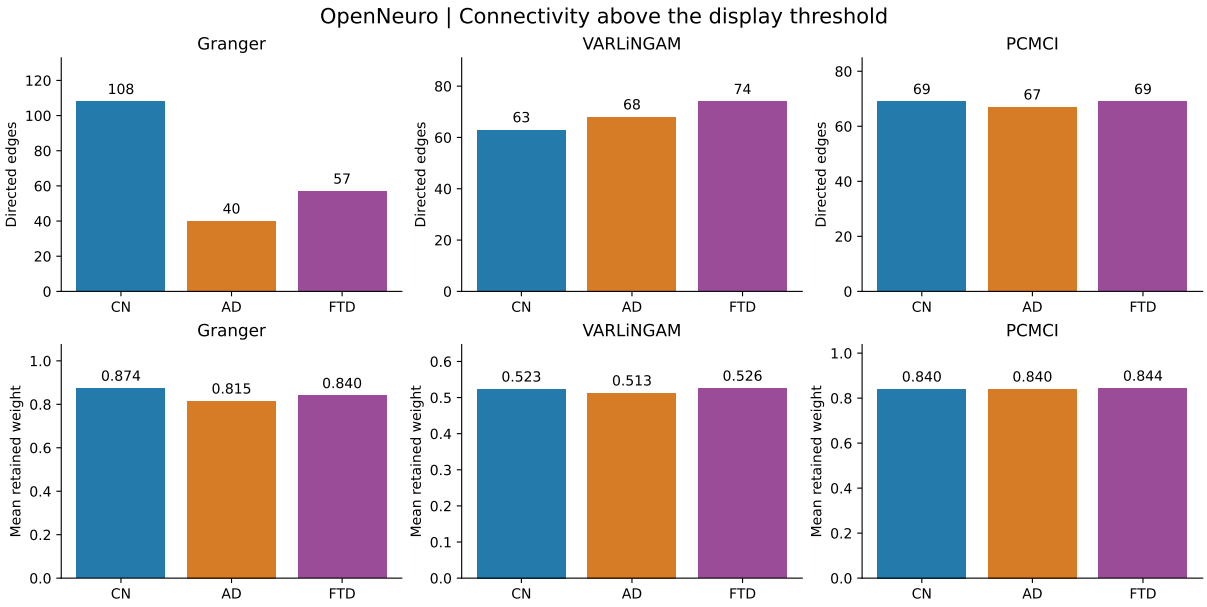


Figure 3: OpenNeuro: edge counts and mean retained weights above the dataset- and method-specific 80th percentile.

Method	Class	$q_{d,m}$	Edges	Density	Mean weight
Granger	CN	0.740	108	0.316	0.874
Granger	AD	0.740	40	0.117	0.815
Granger	FTD	0.740	57	0.167	0.840
VARLiNGAM	CN	0.313	63	0.184	0.523
VARLiNGAM	AD	0.313	68	0.199	0.513
VARLiNGAM	FTD	0.313	74	0.216	0.526
PCMCI	CN	0.790	69	0.202	0.840
PCMCI	AD	0.790	67	0.196	0.840
PCMCI	FTD	0.790	69	0.202	0.844

Table 2: OpenNeuro: thresholded class-average connectivity statistics.

5.2 BrainLat

This dataset contains 85 subjects (32 CN, 35 AD and 18 FTD), with 28 retained epochs per subject.

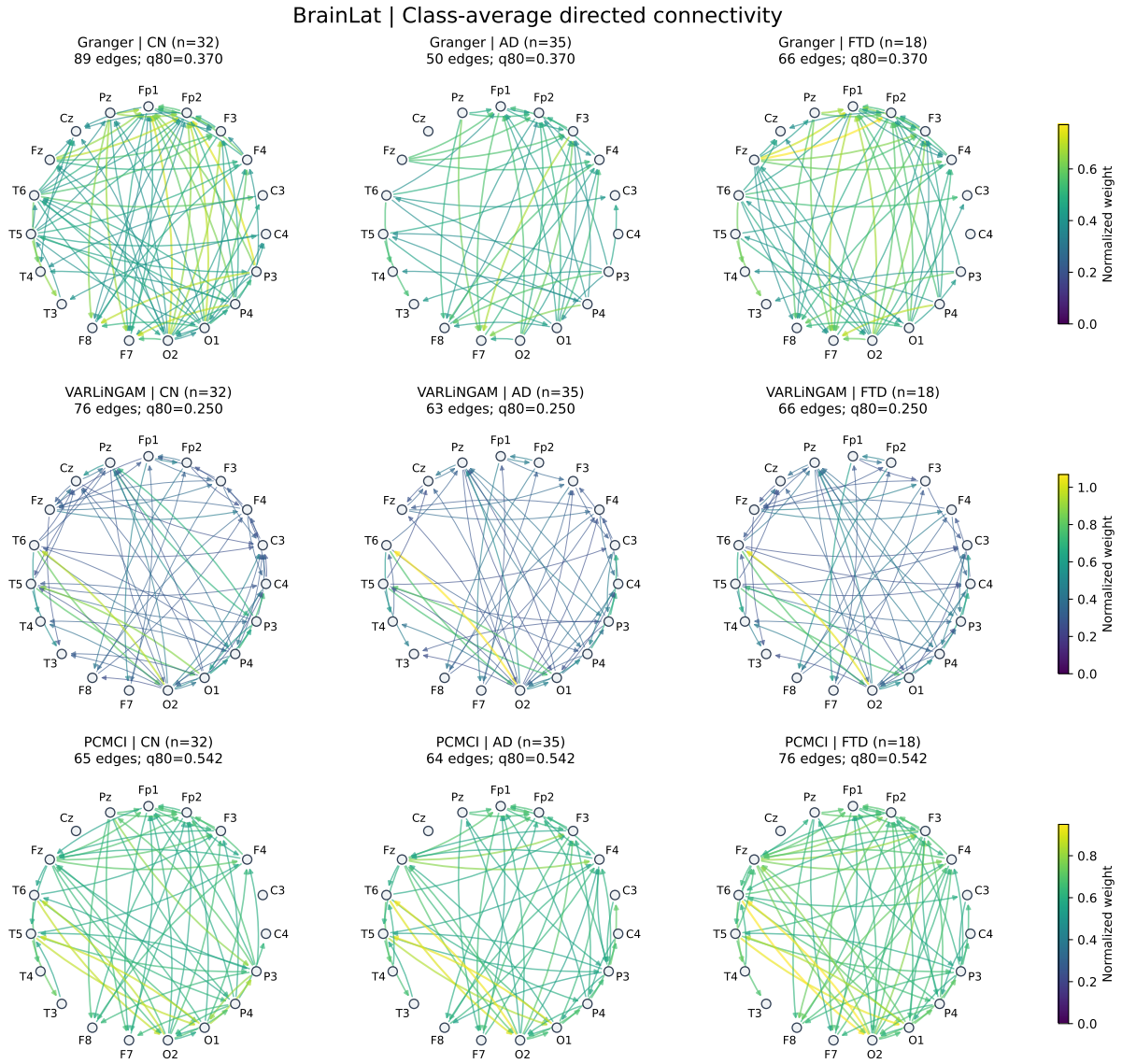


Figure 4: BrainLat: class-average directed networks. Arrows run from source to target; color and width encode normalized weight. Circular positions indicate channel order, not anatomy. Scales and thresholds are shared across classes within each method.

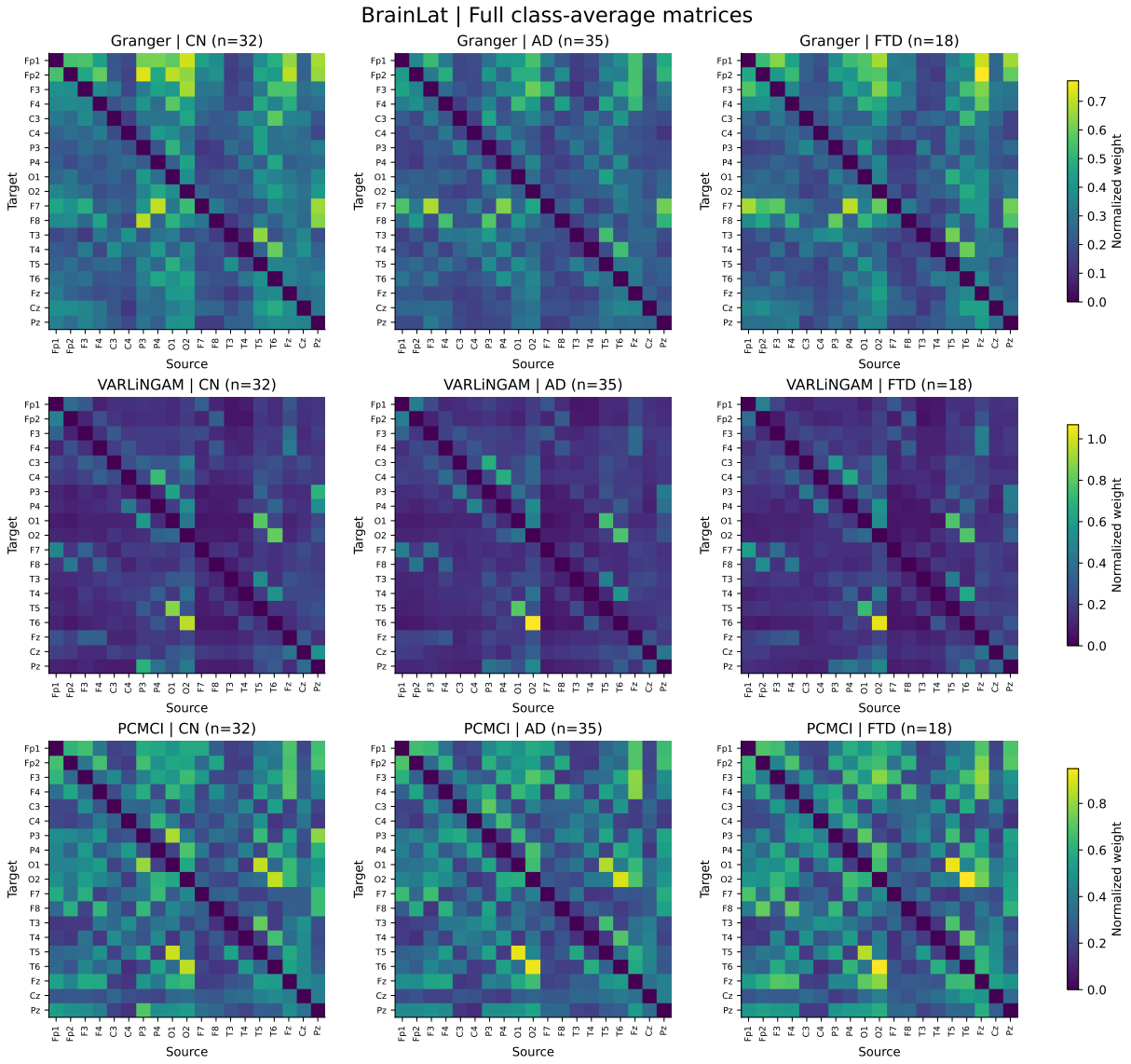


Figure 5: BrainLat: unthresholded class-average matrices. Rows are targets, columns are sources; each method shares its color scale across classes.

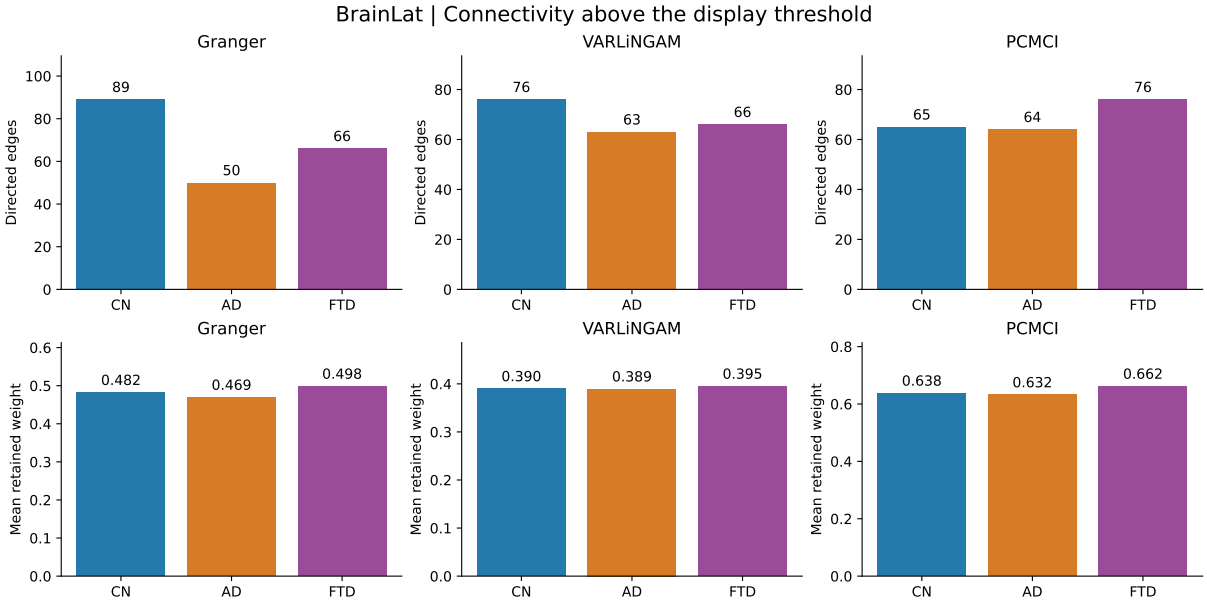


Figure 6: BrainLat: edge counts and mean retained weights above the dataset- and method-specific 80th percentile.

Method	Class	$q_{d,m}$	Edges	Density	Mean weight
Granger	CN	0.370	89	0.260	0.482
Granger	AD	0.370	50	0.146	0.469
Granger	FTD	0.370	66	0.193	0.498
VARLiNGAM	CN	0.250	76	0.222	0.390
VARLiNGAM	AD	0.250	63	0.184	0.389
VARLiNGAM	FTD	0.250	66	0.193	0.395
PCMCI	CN	0.542	65	0.190	0.638
PCMCI	AD	0.542	64	0.187	0.632
PCMCI	FTD	0.542	76	0.222	0.662

Table 3: BrainLat: thresholded class-average connectivity statistics.

The circular plots show the identity and direction of the strongest displayed links; the heatmaps retain weaker connections that the network display omits. In the updated OpenNeuro summary, Granger shows the clearest class separation in displayed edge count: CN retains 108 directed edges above the shared OpenNeuro Granger threshold, compared with 40 for AD and 57 for FTD. VARLiNGAM is more even and increases mildly from CN to FTD (63, 68 and 74 edges), while PCMCI is almost class-balanced in retained edge count (69, 67 and 69 edges). Mean retained weights vary much less than edge counts within each OpenNeuro method, especially for PCMCI.

In BrainLat, the updated plots show a similar Granger ordering in which CN has the largest displayed edge count (89), AD the smallest (50), and FTD is intermediate (66). VARLiNGAM again varies moderately across classes, with CN retaining 76 edges and AD/FTD retaining 63 and 66. PCMCI differs from this pattern: FTD has the largest displayed count (76) and also the largest mean retained weight (0.662). Across both datasets, VARLiNGAM has lower numerical mean retained weights than Granger or PCMCI because its saved values are normalized structural coefficient magnitudes rather than transformed p-value evidence or partial-correlation magnitudes.

A larger retained-edge count does not establish a disease effect. Group comparisons would require subject-level inference, multiplicity control and sensitivity analysis across thresholds. The different estimators have distinct weight definitions and should not be compared as if they shared raw units.

The updated figures are therefore descriptive summaries of the exported graph representations, not inferential tests of clinical group differences.

6 Three-Class MLP Classification

6.1 Experimental setup

The MLP takes 494 inputs: the flattened 19×8 node-feature matrix (152 values) concatenated with the 342 off-diagonal directed adjacency values. A separate model is trained for each connectivity method and dataset. Its architecture is

$$\text{Linear}(494, 32) \to \text{ReLU} \to \text{Dropout}(0.1) \to \text{Linear}(32, 3).$$

The output class order is CN, AD, FTD. Subject-level splits use nominal 70/10/20 percent training/validation/test fractions with per-class integer rounding and seed 42 in the saved runs. The three methods share exactly the same subject partition within each dataset. The classifier creates its own splits rather than reading the preprocessing export's `splits.json`.

Standardization is fitted only on training subjects. Training uses full-batch AdamW, learning rate 0.005, weight decay 0.0001 and cross-entropy weighted by inverse training-class frequency. Validation uses unweighted cross-entropy. Runs allow 500 epochs, with patience 50; the weights with the lowest validation loss are restored. The saved runs use PyTorch `2.9.1+rocm7.2.1` with one thread, as recorded in the dataset-specific `run_config.json` files. The results below are taken from the dataset-specific `results/mlp/openneuro/classifier_results.json` and `results/mlp/brainlat/classifier_results.json` files. Each file describes an independent within-dataset experiment.

For confusion matrix C, with true classes on rows,

$$\text{Accuracy} = \frac{\sum_c C_{cc}}{\sum_{c,k} C_{ck}}, \qquad \text{BA} = \frac{1}{3} \sum_c \frac{C_{cc}}{\sum_k C_{ck}}.$$

All three classes are present in each test split. Balanced accuracy gives equal weight to their recalls; uniform-chance balanced accuracy is 1/3.

6.2 OpenNeuro: independent within-dataset evaluation

The split contains 60 training, 9 validation and 18 test subjects. Test support is 6 CN, 7 AD and 5 FTD.

Adjacency	Balanced accuracy	Accuracy	Best epoch	Epochs run
Granger	0.432	0.444	16	66
VARLiNGAM	0.605	0.611	20	70
PCMCI	0.443	0.444	4	54

Table 4: OpenNeuro: saved MLP test results, seed 42.

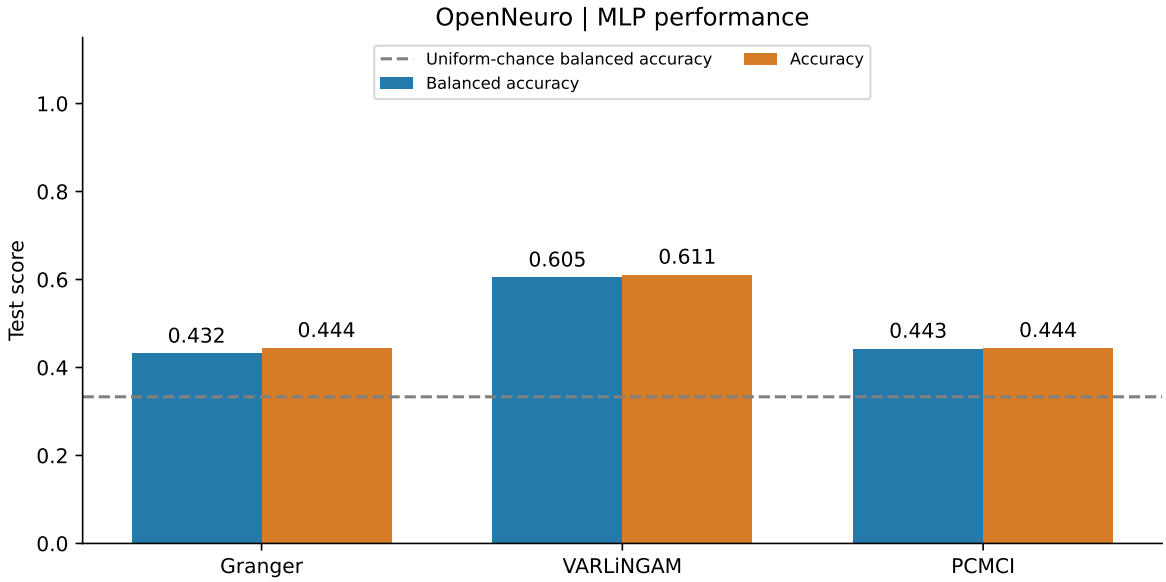


Figure 7: OpenNeuro: MLP test scores by connectivity method.

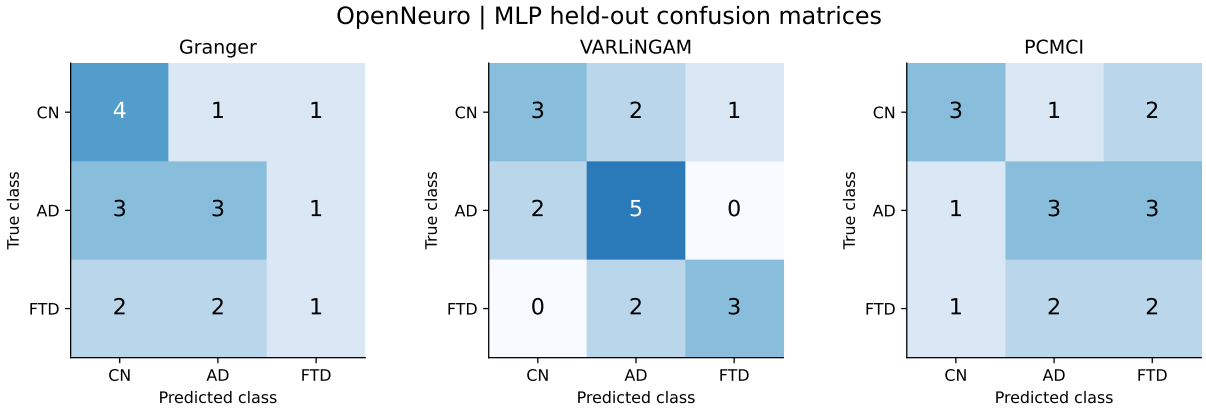


Figure 8: OpenNeuro: test confusion matrices; true classes on rows and predictions on columns.

6.3 BrainLat: independent within-dataset evaluation

The split contains 59 training, 8 validation and 18 test subjects. Test support is 7 CN, 7 AD and 4 FTD.

Adjacency	Balanced accuracy	Accuracy	Best epoch	Epochs run
Granger	0.821	0.833	6	56
VARLiNGAM	0.821	0.833	14	64
PCMCI	0.607	0.667	9	59

Table 5: BrainLat: saved MLP test results, seed 42.

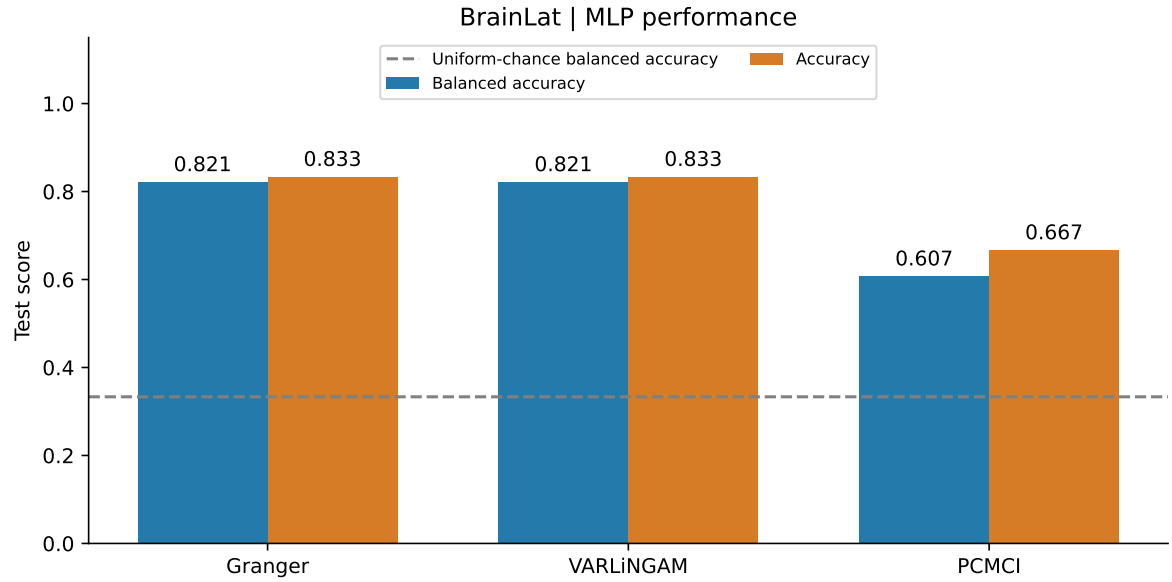


Figure 9: BrainLat: MLP test scores by connectivity method.

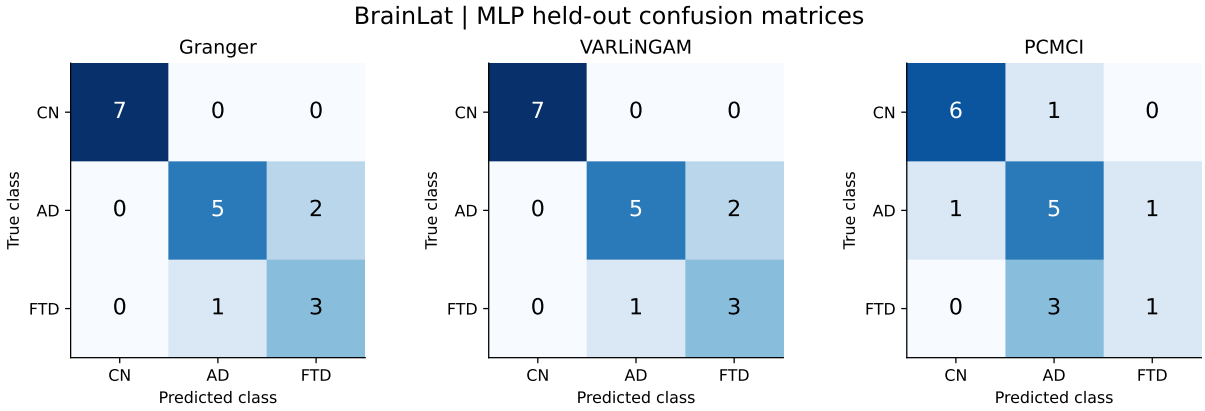


Figure 10: BrainLat: test confusion matrices; true classes on rows and predictions on columns.

6.4 Interpretation and scope

OpenNeuro. The independent run uses seed 42. Granger obtains balanced accuracy 0.432 and accuracy 0.444 (8 of 18 correct). CN, AD and FTD recalls are 0.667, 0.429 and 0.200. The best validation loss is 0.503 at epoch 16 of 66 executed epochs. VARLiNGAM obtains balanced accuracy 0.605 and accuracy 0.611 (11 of 18 correct). CN, AD and FTD recalls are 0.500, 0.714 and 0.600. The best validation loss is 0.421 at epoch 20 of 70 executed epochs. PCMCI obtains balanced accuracy 0.443 and accuracy 0.444 (8 of 18 correct). CN, AD and FTD recalls are 0.500, 0.429 and 0.400. The best validation loss is 0.682 at epoch 4 of 54 executed epochs. On OpenNeuro, VARLiNGAM achieves the highest observed test balanced accuracy (0.605).

The confusion matrices show that VARLiNGAM correctly identifies 3 of 6 CN, 5 of 7 AD and 3 of 5 FTD test subjects. Granger and PCMCI each classify 8 of 18 subjects correctly, but PCMCI has a slightly higher balanced accuracy because the class recalls differ. FTD recall is 0.200 for Granger, 0.600 for VARLiNGAM and 0.400 for PCMCI. All three observed balanced accuracies exceed the uniform-chance reference of 1/3; this comparison alone is not a statistical significance test.

BrainLat. The independent run uses seed 42. Granger obtains balanced accuracy 0.821 and accuracy 0.833 (15 of 18 correct). CN, AD and FTD recalls are 1.000, 0.714 and 0.750. The best validation loss is 0.940 at epoch 6 of 56 executed epochs. VARLiNGAM obtains balanced accuracy 0.821 and accuracy 0.833 (15 of 18 correct). CN, AD and FTD recalls are 1.000, 0.714 and 0.750. The best validation loss is 0.476 at epoch 14 of 64 executed epochs. PCMCI obtains balanced accuracy 0.607 and accuracy 0.667 (12 of 18 correct). CN, AD and FTD recalls are 0.857, 0.714 and 0.250. The best validation loss is 0.758 at epoch 9 of 59 executed epochs. On BrainLat, Granger and VARLiNGAM tie for the highest observed test balanced accuracy (0.821).

Granger and VARLiNGAM produce identical test confusion matrices: both correctly classify all 7 CN subjects, 5 of 7 AD subjects and 3 of 4 FTD subjects. Their three errors consist of two AD subjects predicted as FTD and one FTD subject predicted as AD. PCMCI correctly classifies 12 of 18 subjects, including only 1 of 4 FTD subjects; the remaining three FTD subjects are predicted as AD. The different validation losses of Granger and VARLiNGAM therefore do not imply different held-out classification performance in this run.

These are independent within-dataset experiments, not training on one dataset and testing on the other. The better BrainLat scores do not isolate a dataset effect: cohorts, acquisition conditions, retained durations and test subjects differ. There are only 18 test subjects in each run; one additional correct prediction changes accuracy by 5.6 percentage points. No confidence intervals or paired significance tests are reported, and the observed method ranking is not a validated model-selection result.

The experiments use node features and adjacency jointly. Without feature-only and adjacency-only ablations, they do not quantify the incremental value of connectivity. Repeated subject-level splits and external-dataset testing are needed to assess robustness and transferability.

7 Output Artifacts

Each processed subject directory contains:

- `X.npy`: node-feature matrix with shape 19×8;
- `preprocessed_epochs.npy`: stationarized epochs used for causal estimation;
- six adjacency files, three directed and three symmetric;
- `y.json`: numerical and textual label;
- `metadata.json`: sampling rate, epoch count, channel count, feature count, and quality-control summary.

At dataset level, `summary.csv`, `config.json`, `splits.json`, and `graphs.pkl` provide traceability and reproducibility.

8 Limitations and Future Work

The EEG-to-graph pipeline and classification experiments have several methodological and experimental limitations.

First, the stationarity criterion is strict because it requires agreement between ADF and KPSS. Future ablations should compare the current rule against ADF-only stationarity or a rule that differences only when both tests indicate non-stationarity. Second, each subject is summarized by a static graph even though EEG connectivity is dynamic. Third, the number of subjects is small for training high-capacity models, especially in the FTD class. Fourth, the connectivity estimates

are statistical and may be affected by latent sources, volume conduction, unmodeled nonlinear effects, and multiple testing.

The natural next step is to train graph neural models using the method-specific adjacency matrices, comparing directed and symmetric variants and reporting ablation studies for node features, Granger, VARLiNGAM, and PCMCI. The current MLP results provide a subject-level baseline for that stage.

9 Conclusion

The project represents EEG recordings as subject-level graphs combining spectral and statistical node features with method-specific directed connectivity. Granger causality, VARLiNGAM, and PCMCI provide complementary representations for connectivity analysis and classification. On OpenNeuro, VARLiNGAM achieves the highest observed test balanced accuracy (0.605). On BrainLat, Granger and VARLiNGAM tie for the highest observed test balanced accuracy (0.821). Dataset-specific connectivity figures preserve the distinction between the two cohorts. The results remain preliminary and require validation across repeated splits and independent cohorts.